

AYUSHMAN BHARAT DIGITAL MISSION

[Update Mobile via API in Healthcare Professional Registry \(HPR\) via APIs](#)

Table of Contents

A. Terminology	3.
B. Get HPR Token	6
1. Login Via Password	6
2. Login Via Mobile	7
3. Login Via Aadhaar	10
C. Generate Verification Mobile OTP	12
D. Re-Generate Verification Mobile OTP	13
E. Verify Mobile OTP	14

A. Terminology

- ABDM - Ayushman Bharat Digital Mission
- **Authorization token** - To obtain, the HRP can call 'sessions API' along with the received ClientID and Client Secret. Same must be used in header for other APIs.
- **Client ID** - Post internal approval, Integrator will receive Client ID to access ABDM APIs
- **Client Secret** - Post internal approval, Integrator will receive Client Secret to access ABDM APIs
- HPRID — Healthcare Professional ID
- Sandbox - Sandbox is a framework that will allow technologies or products to be tested in the contained environment in compliance with ABDM standards.

Verify Mobile Via API for registered health professional

Below document explains how to update professional's details in HPR systems using APIs.

List of APIs for Upload document in HPR

1	HPRID Gateway Token	
2	Get HPR Token	
3	Generate Verification Mobile - OTP	
4	Re-Generate Verification Mobile - OTP	
5	Verify Mobile - OTP	

Step 1. HPRID Gateway Token

Authentication Token need be generated using client ID and client Secret. To obtain clientId and clientsecret, please register on [Sandbox Application](#).

Headers

REQUEST-ID :

Unique UUID for tracking the end-to-end unique transaction.

Code example: java

<https://www.uuidgenerator.net/dev-corner/java>

TIMESTAMP:

The actual time when the request was initiated, ISO 8601 represents the date and time by starting with the year, followed by the month, the day, the hour, the minutes, seconds, and milliseconds

X-CM-ID :

This value depends on the environment where API is getting executed.

Eg. SBX or PROD.

For SBX- sbx and for PROD- abdm

Method	POST
URL	https://dev.abdm.gov.in/api/hiecm/gateway/v3/sessions
Request Body	<pre>{ "clientId": "*****", "clientSecret": "3***bf3b-d**d-47a2-be2f-10d*****1d9", "grantType": "client_credentials" }</pre>

Response – Previously generated Transaction ID will be returned in the response

Sample Response :

```
{
  "accessToken": "eyJhbGciOiJSU-
zI1NiIsInR5cCI6IkpXLTUwMzcsImhhdCI6MTczNzNmMzgzNywianRpIjoizWVkbWVjMTZtZTc3ZC00OTMyLTg4N2YtNWRiNTZ
kNWRjNzUyIiwiaXNzI-
joiaHR0cHM6Ly9kZXYubmRobS5nb3YuaW4vYXV0aC9yZWVsbXVjY2VudHJhbC1yZWdp3RyeSIsImF1ZCI6ImFjY291bnQ1LC
```

```

JzdWIio-
iI3YzBkY2ViZi0zNWYwLTQ3OWItYTgyYS0zZWJmNGI5YmUxYWUiLCJ0eXAiOiJCZWZyZXIiLCJhenAiOiJoZnIiLCJzZXNzaW
9uX3N0YXRlIjoInWU2MGZlMjMtNjgzOC00NTMyLTg5NmItNDlkN2JlMjQ5ZWQwIiwiYWNYIjoIMSIsInJlYWxtX2FjY2VzcyI
6eyJyb2xlcYI6WyJoZnIiLCJBQ0NPVU5UX1RBR19BREQiLCJvZmZsaW5lX2FjY2VzcyIsImh1YWx0aElkIiwiaGlwIiwiaHBf
aWQiXX0sInJlc291cmNlX2FjY2VzcyI6eyJhY2NvdW50Ijpb7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYXdllWFjY
291bnQtbGlua3MiLCJ2aWV3LXBjb2ZpbGUiXX19LCJzY29wZSI6Im9wZW5pZCB1bWVpYCBwcm9maWx1Iiw1Y2xpZW50SG9zdC
I6IjEwMC42NS4xNjAuMjE0Iiw1Y2xpZW50SWQiOiJoZnIiLCJlbWVpY2FjY2VzcyI6ZmFsc2UsInByZWZlcnJlZlF91c2V
ybmFtZSI6InNlcnZpY2UtYWNjb3VudC10ZnIiLCJjbGllbnRBZGRyZXNzIjoIMTAwLjY1LjE2MC4yMTQifQ.HOeiaZpKEvZR3
2wGrNM1kDzg0-
9lfyeVIdztTxpMnYT5cY78WN8kBI5t1Es6UM0o5n3e2WbvRy2ronljwblKGie_eBhfyv0qWTOwnQL4wAHFLc2zK8in1wx6QF
72K5xVbQWIWv8rHQmyyGActj47w7BXYahcNE4_iCc3PATLpMj-UnNUJ2PCdQfCk1AUKrK3az5-GpkWbepE_FdRVn8LT-
340Qqcxyf7rFlNF7tw2hS15jgk8aX0YTzDEVxdVQ9jWvc2Nq91Bol8g9ybBgn_srNvLdi8xe0SG_5ATYVN385nr_Vi13z7Asx
i-0_dkfCpPc4wuVZq0wDcRXgZkB9hA",
    "expiresIn": 1200,
    "refreshExpiresIn": 1800,
    "refreshToken": "eyJhbGciOi-
JIUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICIyMWU5NzA4OS00ZTcxLTQyNGEtOTAzYS1jOTAyMWM1NmFlNWYifQ.eyJl
eHAi-
OjE3MzczNTU2MzcsIm1hdCI6MTczNzM1MzgzNywianRpIjoizjM0NDI1ZDYtNGEyZC00ZGM5LTg5NDEtOTA1YTE4MTYxODE1I
iwiaXNzI-
joiaHR0cHM6Ly9kZXYubmRobS5nb3YuaW4vYXV0aC9yZWZsbXNvY2VudHJhbC1yZWdpc3RyeSIsImF1ZCI6Imh0dHBzOi8vZG
V2Lm5kaG0uZ292Lm1uL2F1dGgvcmlhbmG1zL2N1bnRyYWwtcmVnaXN0cnkiLCJzZWIiOiI3YzBkY2ViZi0zNWYwLTQ3OWItYTg
yYS0zZWJmNGI5YmUxYWUiLCJ0eXAiOiJSZWZyZXNoIiw1YXpwIjoiaGZyIiwic2VzcyI6ZmFsc2VzL2Vbl9zdGF0ZSI6IjVlNjBmZTIzLTY4
MzgtNDUzMi04OTZiLTQ5ZDdiZTI0OWVkcjE0NjB1Ijoib3BlbmlkIGVtYWlsIHByb2ZpbGUifQ.WMCsLMSuczpJqtNSG_
THy88tQ-0DAxrY1U3i-TTqT44",
    "tokenType": "bearer"
}

```

B. Get HPR Token

Users can get HPR Token in three ways.

- i. [Login Via Password API,](#)
- ii. [Login Via Mobile OTP API,](#)
- iii. [Login Via Aadhaar OTP API](#)

Users can get HPR token by choosing any given option

I. Login Via Password

Users need to input HPR id and password to get HPR token.

Add the `accessToken` received from the HPRID Gateway Token API for authorization.

Note: Append tokenType Bearer before the access token, give a space and then add your access token as shown below.

Key: Authorization

Value: Bearer<Leave Space><Your Access Token goes here>

Method	Post
URL	https://apihpsbx.abdm.gov.in/v4/int/api/v1/auth/authPassword
Request Body	<pre>{ "idType": "hpr_id", "domainName": "@hpr.abdm", "hprId": "username@hpr.abdm", "password": "000000" }</pre>

Sample Response

```
{
  "token": "eyJhbGciOiJSUzUxMiJ9eyJzdWIiOiI3MS00MzgZLTZlZDUtMjY4NiIsImNsaWVudElkIjoizGlnaWRvY3RvciIsInN5c3RlbSI6IkhQSUQiLCJocHJlZzE5bWJlciI6IjcxLTQzODMtMTMwNS0yNjg2IiwibW9iaWx1IjoiotTU4MjEyNjk5MmhwcklkIjoibmFyZXNMjBAaH ",
  "expiresIn": 1800,
  "refreshToken": "eyJhbGciOiJSUzUxMiJ9eyJzdWIiOiI3MS00MzgZLTZlZDUtMjY4NiIsImNsaWVudElkIjoizGlnaWRvY3RvciIsInN5c3RlbSI6IkhQSUQiLCJ0eXAiOiJSZWZyZXNoIiwizXhwIjoxNjg5MDgzMzcxcXJpYXQiOjE2ODkwNzI1NzF9.VMhRvuEkTcujrSJOAwccZFeixPRFclIgfWICWmd9WKR8mtQEkJvWfTXYeOT8UAzfV7kEsNXoL7",
  "refreshExpiresIn": 10800
}
```

II. Login Via Mobile OTP

Users need to follow the following steps to get HPR token

1	Send OTP
2	Get Public Certificate
3	HPRID Verify OTP
4	Login with HPRID

1. Send OTP

Add the accessToken received from the HPRID Gateway Token API for authorization.

Note: Append tokenType Bearer before the access token, give a space and then add your access token as shown below.

Key: Authorization

Value: Bearer<Leave Space><Your Access Token goes here>

Method	Post
URL	https://apihpsbx.abdm.gov.in/v4/int/api/v2/auth/loginViaMobileSendOTP
Request Body	<pre>{ "mobile": "9999999999" }</pre>

Sample Response
<pre>"txnId": "061c660d-8752-4639-945d-e77a7ea6f564", "mobileNumber": null</pre>

2. Get Public Certificate

Users need to get public certificate to encrypt the OTP.

Method	Get
URL	https://apihspsbx.abdm.gov.in/v4/int/api/v1/auth/cert

Sample Response

```
-----BEGIN PUBLIC KEY-----
MIIBljANBgqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA6avEoNMbZDtjs7RHc1Si
hYeLvGPYV76mh2ncU2FAyCkK3ubE/7UE2HkimL4ftAqFN3fcKv8430oV3gAuvrk/
5slOoG8NOMrVflnakdFZyDae66MHVuc1Rzrv8fKDRFRz8UTa1S2xZ9uXRCieQCeB
xeHmDtgcyGDfQzqcTOLGCPu6tLWda/QhW7zeUpowglIBknhlf2nf916lsM1fWnO7
qVUBh6E3PXoLrW4NtvOwEMh3V62BNGzNmW3u8hxidp+uSe3Us/t4WFJRURnAhO5
IDAzojqkWYHkYZTDvidLBvfoX7ZbBPz03u42AOm/uDu4gG/oYqeB5zXe7qJ1ecLu SQIDAQAB
-----END PUBLIC KEY-----
```

3. HPRID verify OTP

After receiving public certificate user needs to encrypt OTP with the public key using RSA Encryption Algo RS512. RSA Encryption to encrypt the data. Cipher Type - RSA/ECB/PKCS1Padding. online tool to encrypt data [click here](#)

Add the accessToken received from the HPRID Gateway Token API for authorization.

Note: Append tokenType Bearer before the access token, give a space and then add your access token as shown below.

Key: Authorization

Value: Bearer<Leave Space><Your Access Token goes here>

Method	Post
URL	https://apihspsbx.abdm.gov.in/v4/int/api/v2/auth/loginViaMobileSendOTP
Request Body	<pre>{ "otp": "1W/rIffFrI/wOeGml7BhfAGu20To1EQ7HowREwXVIKAnXCyeY31KMDUfI/VlQoHs9ocbwvG4rBCC Xof5jcsHsCs4ZntqI6PYle3Ps5IsGH+7sPO8u9SLMy5x9xBoI3VqnrUARxBb7o", "txnId": "fc6d879c-e535-4ff1-9443-5dc6031efcc1" }</pre>

Sample Response

```
{
  "txnId": "fc6d879c-e535-4ff1-9443-5dc6031efcc1",
  "mobileLinkedHpIdDTO": [
    {
      "hprIdNumber": "***-****-0326-3829",
      "name": "Test",
      "hprId": "*****@hpr.abdm"
    }
  ]
}
```

4. Login with HPRID

Users need to pass hprIdNumber and txnId received in the **HPRID verify OTP** Api. Add the accessToken received from the HPRID Gateway Token API for authorization.

Note: Append tokenType Bearer before the access token, give a space and then add your access token as shown below.

Key: Authorization

Value: Bearer<Leave Space><Your Access Token goes here>

Method	Post
URL	https://apihspsbx.abdm.gov.in/v4/int/api/v2/auth/login/userAuthorizedToken
Request Body	{ "hpld": "***-****-0326-3829", "txnld": "fc6d879c-e535-4ff1-9443-5dc6031efcc1" }

Sample Response

```
{
  "token": "eyJhbGciOiJSU-
zUxMiJ9eyJzdWIiOiI3MS0xMzc0LTAzMjYtMzgyOSIsImNsaWVudEIkIjoizGInaWRvY3RviIsInN5c3RlbS161khQSUQiLCJ0eXA-
iOiJSZWZyZXNoli-
wizXhw1joxNjg5NTkwMz15LCJpYXQiOiJE2ODk1Nzk1Mj19.SVxOjWXYk2LVcD5t3wRYgBlAMxyjVMLjJ0MGtv9ZPpVOBWJUPpwZk8
q6KikP8ZPas",
  "expiresIn": 1800,
  "refreshToken": "eyJhbGciOiJSU-
zUxMiJ9eyJzdWIiOiI3MS0xMzc0LTAzMjYtMzgyOSIsImNsaWVudEIkIjoizGInaWRvY3RviIsInN5c3RlbS161khQSUQiLCJ0eXA-
iOiJSZWZyZXNoli-
wizXhw1joxNjg5NTkwMz15LCJpYXQiOiJE2ODk1Nzk1Mj19.SVxOjWXYk2LVcD5t3wRYgBlAMxyjVMLjJ0MGtv9ZPpVOBWJUPpwZk8
q6KikP8ZPas ",
  "refreshExpiresIn": 10800
}
```

III. Login Via Aadhaar

For login via Aadhaar OTP you need to follow the following steps.

1	Send OTP
2	Verify OTP

1. Send OTP

Add the accessToken received from the HPRID Gateway Token API for authorization.

Note: Append tokenType Bearer before the access token, give a space and then add your access token as shown below.

Key: Authorization

Value: Bearer<Leave Space><Your Access Token goes here>

Method	Post
URL	https://apihpsbx.abdm.gov.in/v4/int/api/v1/auth/init
Request Body	<pre>{ "idType": "hpr id", "domainName": "@hpr.abdm", "authMethod": "AADHAAR OTP", "hprId": "test@hpr.abdm" }</pre>

Sample Response

```
"txnId": "061c660d-8752-4639-945d-e77a7ea6f564",
"mobileNumber": null
```

2. Verify OTP

Users need to input otp and txnId from above API Send Aadhaar OTP.

Add the `accessToken` received from the HPRID Gateway Token API for authorization.

Note: Append tokenType Bearer before the access token, give a space and then add your access token as shown below.

Key: Authorization

Value: Bearer<Leave Space><Your Access Token goes here>

Method	Post
URL	https://apihpsbx.abdm.gov.in/v4/int/api/v1/auth/confirmWithAadhaarOtp
Request Body	<pre>{ "otp": "308709", "txnId": "effbf557-9f2b-40f7-a8c1-9658a08bf410" }</pre>

Sample Response

```
{
  "token": "eyJhbGciOiJSU-  
zUMi9eyJzdWIiOiI3MS0xNjg3LTI3NTUtNzg1MSIsImNsaWVudElkIjoizGlnaWRvY3RvcisiInN5c3RlbSI6IkhQSUQiLCJ0eXAi  
OiJSZWZyZXNoIiwiaXhwIjoxNjg5NjA0OTclLCJpYXQiojE2ODk1OTQxNzV9. ",
  "expiresIn": 1800,
  "refreshToken": "eyJhbGciOiJSU-  
zUMi9eyJzdWIiOiI3MS0xNjg3LTI3NTUtNzg1MSIsImNsaWVudElkIjoizGlnaWRvY3RvcisiInN5c3RlbSI6IkhQSUQiLCJ0eXAi  
OiJSZWZyZXNoIiwiaXhwIjoxNjg5NjA0OTclLCJpYXQiojE2ODk1OTQxNzV9. ",
  "refreshExpiresIn": 10800
}
```

c. Generate Mobile - OTP

In this API's system will send a 6-digit OTP to user input mobile number to verify user identity.

Add the accessToken received from the HPRID Gateway Token API for authorization.

Note: Append tokenType Bearer before the access token, give a space and then add your access token as shown below.

Key: Authorization

Value: Bearer<Leave Space><Your Access Token goes here>

Method	Post
URL	https://apihspsbx.abdm.gov.in/v4/int/apis/v1/doctors/generate-mobile-otp
Request Body	<pre>{ "hpr_token": "eyJhbGciOiJSUzUxMiJ9.eyJzdWIiOiI3MS00MzgZLTE ", "officialMobile": "PFIi0AQJkAKIdLL6ytdJLKFObqXab9rwTvOEPHflksQBkMzIIb8HMDYUzZfn3AUdefedSbb+B4sPwi 72lsaaNRkpXygWRF0GWntEwD/GL9t3z9N07ELWYb9hcaHDZccyxIYW4cwR4DgfrbCHdvlEj14eJ1hJH LuctGPE5OFxkEP/WL80JbXaW9DJkwPpDEzQpMYKKT17iCTp7pQer8337NZofO1D1aYiDfEnA9E1HMTy PCGFjvmbcL32hNqGsgpHKYNh4rHXCo4RwP5UQKWDYI1jLZqbWLp0a9GQu9nC1hyP5IR5LRCASzvhiRf rRk+Y660xuDSCvOKb+uUPcN3ZFA== " }</pre>

Sample Response

```
"status": "success",  
"txnId": "3e8457d4-d185-4b8c-b359-3b7f36e27d80"
```

Response Body Parameters

Field Name	Mandatory	Datatype	Size	Comments
hpr_token	Yes	Varchar		Token you will received in get HPR token
officialMobile	Yes	Number	10	

D. Re-Generate Mobile OTP

In this API's if user didn't receive an OTP after this user will use this api to regenerate OTP again for mobile verification.

Add the accessToken received from the HPRID Gateway Token API for authorization.

Note: Append tokenType Bearer before the access token, give a space and then add your access token as shown below.

Key: Authorization

Value: Bearer<Leave Space><Your Access Token goes here>

Method	POST
URL	https://apihpsbx.abdm.gov.in/v4/int/apis/v1/doctors/regenerate-mobile-otp
Request	<pre>{ "hpr_token ": "eyJhbGciOiJSUzUxMiJ9", "officialMobile": "PFiI0AQJkAKIdLL6ytdJLKFObqXab9rwTvOEPHflksQBkMzIIb8HMDYUzZfN3AUdefedSbb+B4sPwi72l saaNRkpX- ygWRF0GWntEwD/GL9t3z9N07ELWYb9hcaHDZccyXIYW4cwR4DgfrbCHdvlEj14eJ1hJHLuctGPE5OFxkEP /WL80JbXaW9DJkwPpDEzQpMYKKT17iCTp7pQer8337NZofO1D1aYiDfEnA9E1HMTyPCGFjvmbcL32hNqGs gpHKYNh4rHXCo4RwP5UQKWDYI1jLZqbWlp0a9GQu9nC1hyP5IR5LRCASzvhiRfrRk+Y660xuDSCvOKb+uU PcN3ZF== " }</pre>

Sample Response
<pre>"status": "success", "txnId": "3e8457d4-d185-4b8c-b359-3b7f36e27d80"</pre>

Field Name	Mandatory	Data types	Size	Comments
hpr token	Yes	String		
officialMobile	Yes	String		

E. Verify Mobile-OTP

In this API's user will update mobile by input OTP along with other requested parameters mentioned in the table explaining API which user receive in Generate Mobile OTP and Re-Generate Mobile OTP.

Add the accessToken received from the HPRID Gateway Token API for authorization.

Note: Append tokenType Bearer before the access token, give a space and then add your access token as shown below.

Key: Authorization

Value: Bearer<Leave Space><Your Access Token goes here>

Method	POST
URL	https://apihpsbx.abdm.gov.in/v4/int/apis/v1/doctors/verify-mobile-otp
Request Body	<pre>{ "hpr_token": "eyJhbGciOiJSUzUxMiJ9", "txnld": "386d9909-c825-4b6b-bb61-505324a6a6f", "otp": "INEz+jUe1GyLpDAGYo6zm8uMRmF+NNs1NKCXpQN8AOswpu3IBBHb1dn0OZ1FcXq7+qSiK5Dcc9iWN9au G2rDYN2M0/jUjeTWffurspwRpQ8CCYK2tSIeD0GaGj07+akqdngmisqmQz6zKqdsbHcmlEDv+aODJIiw/ 6sGwTby0jQBcNObrW1BrbPlrkBP/jQP0lnJiUzeesNKwfJp7YEVN+5CFcOQcBiZA3bL7Y5zLjnxwuX6lM Q8HWZuJluXtXldQN1TtgPa9/ewDjnsN5T/i/02c128NghWmxOlJwxT18tzeO6PUgjRfxXCFwDbgVwvxTw PgPAU8HSGqHyX6zEoFw==" }</pre>

Sample Response

```
"status": "success",  
"msg": "Success"
```

Request body parameters.

Field Name	Mandatory	Data types	Size	Comments
hpr_token	Yes	String		Token you received in GET hpr token
txnld	Yes	String		You received response of Generate Mobile OTP and regenerate mobile OTP
Otp	Yes	Number	6	

